Check for updates

Received: 21 March 2025 | Accepted: 2 September 2025

DOI: 10.1111/age.70044

REVIEW

ANIMAL GENETICS WILEY

Decoding DNA methylation in *Staphylococcus aureus* mastitis: Implications for immune regulation and disease resistance

Apeksha[1] | Abhishek Mahendra Todkari[2] | Ashok Chaudhary[1] | Mir Mehroz Hassan[1] | Diksha Upreti[3] | Shri Ram Saini[1] | Sheikh Firdous Ahmad[1,4] | Ashwni Kumar Pandey[1]

[1]Division of Animal Genetics, ICAR-Indian Veterinary Research Institute, Izatnagar, Uttar Pradesh, India

[2]Department of Animal Husbandry, Government of Maharashtra, Satara, Maharashtra, India

[3]Division of Animal Reproduction, ICAR-Indian Veterinary Research Institute, Izatnagar, Uttar Pradesh, India

[4]Livestock Production and Management Section, ICAR-Indian Veterinary Research Institute, Izatnagar, Uttar Pradesh, India

Correspondence
Ashwni Kumar Pandey, Division of Animal Genetics, ICAR-Indian Veterinary Research Institute, Izatnagar, Uttar Pradesh, India.
Email: ashwni.pandey@gmail.com

Abstract

Mastitis is a major health and economic threat to the dairy industry, causing massive losses every year worldwide. *Staphylococcus aureus* is the chief pathogen, responsible for most of the subclinical as well as a considerable portion of the clinical cases. Conventional control strategies, including antibiotic treatment and genetic selection for resistance, have shown limited success due to antimicrobial resistance and low heritability of mastitis resilience traits. This calls for the exploration of novel approaches such as epigenetics, that offers insights into host–pathogen interactions beyond the genetic variations. This review focuses on DNA methylation changes in the mammary gland that occur during *S. aureus* mastitis. Recent research has identified immune suppression and pathogen persistence in relation with DNA methylation during the disease. The microbe has been reported to alter the methylation status of regulatory regions for many immune genes such as *CXCR1*, *TNF-α*, *IL6R*, *IL10*, and *C3*, resulting in dysregulation of immune responses in the host, thereby facilitating pathogen persistence and chronic infection. Along with its own virulence factors, differential DNA methylation status of such genes during infection helps the pathogen to escape host defence, and decreases the intensity of inflammation. Thus, understanding these mechanisms can open new avenues in the field of disease diagnosis, animal selection, and immunotherapy among others. Such an integrative approach offers a revolution in mastitis control strategies, ensuring better health and productivity in dairy animals. © 2025 Stichting International Foundation for Animal Genetics.

KEYWORDS
DNA methylation, epigenetics, immune response, mastitis, Staphylococcus aureus

INTRODUCTION

The global demand for milk and dairy products has increased significantly over the past few decades, fuelled by factors such as population growth, urbanisation, rising incomes, and evolving dietary preferences (Cheng et al., 2022; Yousefian et al., 2024). In response, the dairy industry has implemented advanced strategies for enhancing per animal milk yield, such as those related to nutrition and management, and genetic selection (Cobirka et al., 2020). The efforts have converged into significant improvements; however, persistent challenges, particularly mastitis, continue to cause substantial losses to the sector. Mastitis is a disease of high concern due to its profound economic, animal welfare, productivity, and public health implications (Asfaw & Negash, 2017; Dalanezi et al., 2020; Detilleux et al., 2015; Narayana et al., 2021). Being a multifactorial condition, it arises from a complex interplay of pathogenic organisms as well as environmental factors, and managemental

© 2025 Stichting International Foundation for Animal Genetics.

practices (Jones & Bailey, 2009). Various pathogens cause the infection, among which *Staphylococcus aureus* has emerged as a major causative agent of both clinical and subclinical mastitis cases, particularly the latter (Reyher et al., 2011). It is widely accepted as a major threat worldwide owing to its ability to evade host immune responses, produce persistent infections, and develop antimicrobial resistance (AMR). The traditional mastitis control strategies have not completely been effective for eradication of the disease, especially due to increasing cases of AMR, and the low heritability of mastitis resilience traits (Chandrasekaran et al., 2014; De Vliegher et al., 2012). This has necessitated the exploration of novel mechanisms regulating disease susceptibility and immune responses. Over the past few years, epigenetic mechanisms, specifically DNA methylation, have drawn the attention of the scientific community as a contributor to shaping host–pathogen interactions and associated immune functions. DNA methylation is a heritable modification in the cytosine–phosphate–guanine (CpG) regions of the genome by addition of a methyl group that influences expression of genes without altering the genetic code itself. There is emerging evidence on the manipulation of the host's DNA methylation levels upon infection by *S. aureus* within the mammary tissues. Specific changes have been reported in immune-related genes and linked to alterations in the immune responses during course of the disease (Dong et al., 2021; Wang et al., 2024). Understanding such modifications offers new insights into mechanisms of mastitis susceptibility and resistance that can be used as novel strategies in control of the disease.

This review aims to inspect the various aspects of DNA methylation-associated changes in the immune responses to *S. aureus*-induced mastitis. We discuss the mechanisms of DNA methylation, the interaction of *S. aureus* with the mammary gland methylome, and its impact on the host immune mechanisms. We highlight important modifications and their effect on the expression of genes related to host defence. Finally, possible areas to be explored and probable applications from the understanding of these epigenetic markers are presented. The integration of epigenetics into mastitis research offers significant potential to uncover novel avenues for a sustainable management of the disease and improving dairy productivity.

MASTITIS IN DAIRY ANIMALS

Mastitis is a complex, production-related disease that is defined as the inflammation of mammary glands. The disease is one of the chief factors that negatively affect quality and quantity of milk in dairy animals, accounting for the majority of culling in cattle and buffalo. It is considered as the most critical economic and health challenge in dairy herds (Cobirka et al., 2020), and is among the most widespread and most costly diseases of dairy animals (Benić et al., 2018; Bogni et al., 2011; Jamali et al., 2018; Narayana et al., 2021). Reduced milk quantity and discarding of poor-quality milk, increased culling from the herd, and additional expenditures on treatment of mastitis lead to huge economic losses to producers (Detilleux et al., 2015; Holmberg et al., 2012). Mastitis poses a significant risk to both food safety and environmental health. Apart from this, an individual animal's reproductive performance (Dalanezi et al., 2020; Kumar et al., 2017) and overall health may deteriorate due to an increased susceptibility to other diseases. Due to the severe pain caused and impact on animal health, mastitis poses a challenge for animal welfare as well (Asfaw & Negash, 2017; Logue & Nolan, 1998). Therefore, it has drawn attention of researchers worldwide over decades, and continues to offer newer avenues in animal production yet to be explored.

The disease can be triggered by physical injury and/or pathological microorganisms, along with contributors such as poor milking practices and environmental factors (Jones & Bailey, 2009). It is classified on the basis of causative agent as environmental or contagious, and as clinical or subclinical based on symptoms manifested in host. Subclinical cases amount for a significantly higher proportion than that of clinical mastitis, respectively, being around 25–65% and 5% (Bathla et al., 2020; Dufour et al., 2012). Subclinical mastitis is attributed to higher financial losses than clinical mastitis (Azooz et al., 2020; Morales-Ubaldo et al., 2023). The severity of mastitis depends on invasive microbes that are involved and the subsequent immune responses produced (Gogoi-Tiwari et al., 2017; Pighetti & Elliott, 2011). There is variability in susceptibility to the same pathogen among cows, and lactation stage also influences the incidence of the infection (Burvenich et al., 2000). The pace, strength, and duration of the immune reaction as well as susceptibility to mastitis are decisively linked to the genetic makeup of an animal (Bobbo et al., 2019; Pighetti & Elliott, 2011).

MASTITIS AGENTS IN BOVINES AND HEGEMONY OF *S. AUREUS*

Mastitis is a multi-aetiological disease involving bacteria, fungi, mycoplasma, viruses, algae, and yeasts, which may have primary or secondary roles in its occurrence, and may either be environmental or contagious. Bacteria are the most important causal agents among these (Krishnamoorthy et al., 2021), involving both Gram-positive and -negative types. In bovine mastitis, *S. aureus*, *Streptococcus uberis*, and *Escherichia coli* are considered as the major pathogens (Heikkilä et al., 2018; Krishnamoorthy et al., 2021; Saidani et al., 2018; Wellnitz & Bruckmaier, 2012). *S. aureus* and *E. coli* are zoonotic, with the latter being environmental (Zi et al., 2018). Gram-negative pathogens, such as *E. coli*, *Klebsiella*

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

species, and *Pseudomonas* species, cause clinical mastitis more often in herds with a low bulk milk somatic cell count (SCC); and by the contrast, Gram-positive pathogens such as *S. aureus*, *Streptococcus dysgalactiae*, and *Streptococcus agalactiae* cause clinical mastitis more often in herds with a high bulk milk SCC (Barkema et al., 1998).

S. aureus is the most prevalent contagious mastitis pathogen and causes both clinical and subclinical conditions (Levison et al., 2016; Maddela et al., 2024; Nigam, 2015; Singha et al., 2021). The microbe colonises skin of mammary gland and multiplies within the organ, while evading the defence system of host to mostly cause subclinical or chronic mastitis, characterised by a reduced milk yield and an increased SCC in milk (Cheng & Han, 2020; Kibebew, 2017; Sutra & Poutrel, 1994). No other signs of disease appear in such cases apart from this; however, due to the microbial enzymes and toxins, the udder tissue is damaged irreversibly, causing a decreased milk production and persistent infection (Cheng & Han, 2020). This necessitates a prolonged use of antibiotics, which are used inappropriately at times, often leading to development of AMR.

Economic losses related to *S. aureus* clinical and subclinical mastitis result from the decrease in milk production and quality, the expenses on antibiotics and veterinary fees, and the withdrawal period of cows treated with antibiotics (Aghamohammadi et al., 2018; Halasa et al., 2007). Horizontal transfer of antibiotic resistance genes (Lerminiaux & Cameron, 2019), the emergence of livestock-associated methicillin-resistant *S. aureus* strains (García-Álvarez et al., 2011; Holmes & Zadoks, 2011), and the public pressure to reduce antibiotic use in animal production are the few important reasons to find efficient non-antibiotic control measures for this disease. Therefore, breeding for mastitis resistant cattle becomes even more important.

IMMUNITY TO *S. AUREUS* IN MASTITIS

Mastitis resistance depends on the immune response mounted within the mammary gland. Physical barriers constitute the first line of defence in the gland against infection. These include udder and teat morphology, milking speed, innate immunity, along with the mechanical and antimicrobial teat canal barrier, through which bacteria must penetrate to cause intramammary infection (Burvenich et al., 2000; Rainard & Riollet, 2006; Günther & Seyfert, 2018). After establishment of infection, innate and adaptive immunity play their role. Strain-specific immune response has also been reported for *S. aureus*. Some strains cause rapid and intense immune response leading to clearance of organism(s) from milk, and hence associated with mild clinical mastitis, while others cause persistent mastitis due to milder immune responses (Engler et al., 2022). Strain diversity (Sivakumar et al., 2023), multiple virulence factors (Soares et al., 2017), persistent infection (Campos et al., 2022), immune evasion (De Jong et al., 2019), and low cure rates of anti-biotherapies (Erskine et al., 2002) are the critical factors making *S. aureus* an important pathogen in context of mastitis. Chronic and subclinical mastitis caused by *S. aureus* has low response to antibiotics due to intracellular colonisation in epithelium (Erskine et al., 2002).

ROLE OF EPIGENETICS IN BREEDING FOR MASTITIS RESISTANCE

Mastitis can be controlled through management practices and antibiotic treatment of affected animals, although these approaches alone have not been able to eradicate this rampant disease (Wang et al., 2020). Further, AMR has been reported time and again, causing low cure rates, along with posing a public health hazard (Barkema et al., 2006; Chandrasekaran et al., 2014). Mastitis control in dairy animals consumes a massive 80% of the total antibiotics that are used in this industry (Ashraf & Imran, 2020), indicating the predicament of the issue. Underscoring the effective, but short-term and limited gains from the above approaches, genetic selection of animals can offer a parallel and permanent solution. Identification of animals' resilience (or susceptibility) for the disease can help in early intervention through preventive measures and also aid in efficient selection (Bouzeraa et al., 2024). Thousands of genes are involved in the immune response against different pathogens, but animals with better responses can be identified using conventional quantitative genetic principles based on phenotype (Mallard et al., 2015). Heritability of immune responses, both cell-mediated (CMIR) and antibody-mediated (AMIR), is satisfactorily high to enable selection of animals for disease resistance (Hernandez et al., 2006; Thompson-Crispi et al., 2012). However, susceptibility to mastitis has been established as a trait with low heritability (de Haas et al., 2002; De Vliegher et al., 2012; Lush, 1950; Walawski, 1999), which necessitates the use of modern breeding strategies involving genomic selection aided by genome wide association studies, transcriptomic, post-transcriptomic, and epigenetic studies (Wang et al., 2020). Gene transcription is regulated by both genetic mutations and epigenetic modifications (Zhang et al., 2018). Genetic markers such as single nucleotide polymorphisms are frequently being used to assist genomic selection for livestock species such as cattle (Bouquet & Juga, 2013; Obšteter et al., 2021). Epigenetics, defined as the study of heritable changes in gene expression that do not involve alterations to the underlying DNA sequence (Ho & Burggren, 2010), has

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

emerged as a powerful tool in recent years for understanding complex traits such as mastitis in livestock. The epigenome encompassing the entire range of these potentially inheritable modifications across the genome, is contingent on the genetic, environmental, and other non-genetic factors (Bernstein et al., 2007; Ibeagha-Awemu & Khatib, 2017). Therefore, unlike the genome, the epigenome is inordinately dynamic through lifetime of an individual, and also, within an individual, different cells/tissues express unique epigenetic profiles, despite harbouring the same genetic instruction.

Mastitis involves a triadic interaction between the host, pathogen, and environment, wherein the host factors include both its genetic and epigenetic components (Dego, 2020). The epigenetic component is affected by environmental cues, which may be external, such as temperature, chemicals, and pathogens, or internal, such as nutrition status, hormones, and oxidative stress (James et al., 2004; Petronis, 2010; Tando & Matsui, 2023). The epigenetic alterations can result through several mechanisms (epigenetic marks) including DNA methylation, histone modifications, non-coding RNAs (ncRNAs), and chromatin remodelling. Among these, DNA methylation is the most studied and characterised epigenetic mechanism that plays a significant role in regulating mammary gland health and function in livestock (Ibeagha-Awemu & Zhao, 2015; Ivanova et al., 2021).

DNA METHYLATION

DNA methylation involves the covalent addition of methyl groups to the 5′ position of cytosine nucleotides in the DNA, usually at CpG sites (Ibeagha-Awemu & Zhao, 2015; Triantaphyllopoulos et al., 2016). These CpG sites containing an unusually high frequency of cytosine and guanine residues, are present in most of the gene promoters, and are typically devoid of methylation (Illingworth & Bird, 2009). Many CpG sites are present in the bovine genome. DNA methyltransferase enzymes such as DNMT1, DNMT3a, and DNMT3b catalyse DNA methylation, using S-adenosyl-methionine as methyl donor (Miranda & Jones, 2007). Here, DNMT1 functions as a “maintenance” methyltransferase, copying existing methylation patterns after DNA replication, while DNMT3a and DNMT3b act as “de novo” methyltransferases, establishing new methylation patterns during development and differentiation, essentially creating new methylation marks on previously unmethylated DNA (Tajima et al., 2022). The process typically leads to gene silencing by blocking transcription factors from binding to the DNA. Hence, in general, hypermethylation in the regulatory regions of DNA is associated with inactivation of genes, whereas the reverse happens at lower levels of methylation. Conversely, gene expression is said to be increased if the hypermethylation occurs within the body of genes (Langevin & Kelsey, 2013). Insights into DNA methylation patterns can help explore the “black box” of phenotypic variation that has not possibly been explained by the genetic variations (Ibeagha-Awemu & Yu, 2021). Approaches for assessment of DNA methylation may either evaluate variability in single DNA fragments, or the entire genome. Studies on mastitis have used methods such as whole-genome bisulphite sequencing (Wang et al., 2024), methylated DNA immunoprecipitation (MeDIP) (Nayan et al., 2022; Song et al., 2016), bisulphite sequencing (Ząbek et al., 2020), nested bisulphite sequencing PCR (Zhang et al., 2018), fluorescence-labelled methylation-sensitive amplified polymorphism (Wang et al., 2019), and methylation-sensitive restriction-site associated DNA sequencing (Methyl-RAD Seq) (Wang et al., 2020). All of the aforementioned methods target genome-wide methylation patterns. Although not used in any mastitis-related study so far, it is worth noting the examples of techniques evaluating single DNA fragments: methylation-specific PCR, combined bisulphite restriction analysis, pyrosequencing, and methylation-sensitive restriction enzyme PCR.

In dairy cattle, the role of DNA methylation in milk production has been well documented. Studies show that methylation differences between lactating and non-lactating stages influence the synthesis of milk components. DNA methylation at specific genomic loci has been associated with milk yield, composition, and mammary gland function (Wang & Ibeagha-Awemu, 2021). This understanding is crucial for addressing and utilising the information on mastitis-related alterations in the DNA. In general, the global CpG methylation is higher in mastitis-positive animals, and more significant in the regulatory regions of DNA such as promoters, first exons, and first introns (e.g. Wang et al., 2023, 2024). DNA methylation within the first intron during *S. aureus*-induced subclinical mastitis in dairy cows was found to influence immune-related pathways, potentially affecting the cow's ability to respond to the infection (Wang et al., 2022). However, some studies have reported lower overall methylation levels in infected animals compared to healthy ones (e.g. Ju et al., 2020; Nayan et al., 2022).

S. AUREUS MASTITIS: PATHOGENESIS AND DNA METHYLATION

S. aureus infection is established in three phases. The first phase involves adhesion of the organism to host cell and extracellular matrix. This is mediated by adhesion molecules such as ClfA and ClfB. In the second phase, invasion of tissues occurs with the help of fibronectin and fibronectin-binding protein, wherein an overexpression of fibronectin-binding protein leads to higher invasion. Finally in the third phase, evasion of host immune response occurs (Almeida et al., 1996; Campos et al., 2022;

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Middleton, 2008). A differential expression of various virulence factors takes place in the different stages of infection. In *S. aureus*, expression of virulence factors is under the control of global regulators such as the accessory gene regulator, which allows transition from colonisation phase to invasive phase, and the alternative transcriptional sigma factor, which promotes expression of adhesions and biofilm (Novick & Geisinger, 2008). Concomitantly, the pathogen manipulates host immune responses through epigenetic alterations including DNA methylation. Studies indicate that *S. aureus* infection induces hypermethylation in proinflammatory cytokine genes such as *TNF-α* and *IL6R* genes, potentially dampening the immune response and allowing persistent infections (Dong et al., 2021; Wang et al., 2020, 2024; Zhang et al., 2018). In buffalo, hypermethylation of promoters for *PZP* (*pregnancy zone protein*) and *CPAMD8* (*complement 3 and pregnancy zone protein-like*, *a2-macroglobulin domain-containing protein 8*) genes has been observed (Nayan et al., 2022). Both the genes are broad spectrum protease inhibitors and are involved in opsonisation and phagocytosis of pathogens. Thus, their downregulation during mastitis supports persistence of *S. aureus* for longer periods inside the host.

HOST IMMUNE RESPONSE AND DNA METHYLATION ALTERATIONS

The mammary gland relies on both innate and adaptive immunity to combat infections. The innate immune system provides first line of defence through physical barriers, pattern recognition receptors (PRRs), inflammation, complement activation, phagocytosis, lactoferrins, cytokines etc. Bacterial penetration into the udder is physically blocked by the tight sphincters at teat ends in between milkings (Jones & Bailey, 2009). However, these sphincters are relaxed in early dry period and immediately after milking, increasing the susceptibility to mastitis during these periods (Rainard & Riollet, 2006). Secondly, keratin and calcium binding protein layer of the teat canal has bacteriostatic and bactericidal properties (Notcovich, 2021; Smolenski et al., 2015). Mammary epithelial cells have PRRs for the recognition of pathogen-associated molecular patterns (Aitken et al., 2011). PRR–pathogen associated molecular pattern binding results in release of various cytokines and chemokines depending on the PPRs involved, and leads to recruitment of the cells of immune system. Cytokines are signalling chemicals that may either promote inflammation upon encounter of pathogen (proinflammatory cytokines) or restrict the activity of proinflammatory cytokines (anti-inflammatory cytokines). The proinflammatory cytokines are released early in infection, as soon as the mammary epithelial cells identify the components of different mastitic bacteria (Cheng et al., 2020; Lahouassa et al., 2007; Vitenberga-Verza et al., 2022).

CXCR1 (C-X-C motif chemokine receptor 1) is a G-protein coupled receptor specifically involved in triggering of neutrophil chemotaxis and activation via its binding to ligands such as IL-8 (CXCL8) and CXCL6 (Turner et al., 2014). Neutrophils being the primary immune cells in udder infections, make this mechanism an essential part of immune responses to mastitis. However, *S. aureus* has evolved sophisticated mechanisms to modulate these immune mechanisms through DNA methylation changes. For example, increased CpG site methylation in the *CXCR1* gene has been observed in *S. aureus*-infected mammary epithelial cells, impairing the host immune defence, and facilitating bacterial persistence (Dinarello, 2000; Mao et al., 2015; Wang et al., 2020). *CSF2RB*, which encodes for the common receptor β chain of cytokines such as GM-CSF, IL3, and IL5, has been reported as downregulated under promoter hypermethylation after *S. aureus* mastitis in buffalo (Nayan et al., 2022). This suggests that subclinical mastitis reduces immune function by downregulation of these cytokines. Similarly, altered DNA methylation profiles were reported for other genes involved in *S. aureus* induced inflammation, such as *IL6R*, *TNF*, *IL10*, and *IL17*. (Wang et al., 2020; Zhang et al., 2018). However, a cell culture study revealed contrasting results, i.e. DNA hypomethylation-led increase in expression of cytokines *TNF-α*, *IL-1β*, *IL-6*, and *IL-8*, and chemokines *CXCL1* and *CXCL6*, in response to bacterial peptidoglycan and lipoteichoic acid (LTA) (Wu et al., 2020). By complex interactions of such mechanisms, *S. aureus* cell wall components promote subclinical and chronic mastitis due to their weak but persistent inflammatory activity (Eckel & Ametaj, 2016). Methylation and gene expression changes are dependent on the type of pathogen as well as the duration of exposure to the pathogen.

PATHOGEN-DEPENDENT GENE EXPRESSION

mRNA expression of inflammatory cytokines and chemokines such as *TNF-α*, *IL1β*, and *IL8* is more pronounced after stimulation from lipopolysaccharide (LPS), the cell wall component of Gram-negative bacteria, as compared to that from LTA, the constituent of Gram-positive bacterial cell wall. Therefore, *S. aureus* is frequently isolated from subclinical and chronic cases, in contrast to Gram-negative bacteria such as *E. coli* (Wellnitz et al., 2011; Yang et al., 2008). *TNF-α*, an important cytokine for inflammation and immunity, was reported as downregulated due to promoter hypermethylation in cases of LTA-bearing *S. aureus* mastitis (Dong et al., 2021; Wang et al., 2020, 2024; Zhang et al., 2018), whereas mammary epithelial cells showed its hypomethylation and resultant increased expression upon LPS stimulation (Dong et al., 2021). The STAT5-binding lactational enhancer present upstream to the *αS1-casein* gene's promoter is normally

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

hypomethylated in lactating mammary glands. A pathogen-specific epigenetic response occurs in this region, where remethylation leads to suppression of αS1-casein synthesis in case of *E. coli* acute mastitis, contributing to a metabolic shift toward immune defence. In contrast, subclinical *S. aureus* infections do not induce any such changes, allowing continued αS1-casein synthesis, which may contribute to the pathogen's long-term survival in the mammary gland (Vanselow et al., 2006).

EXPOSURE-DEPENDENT GENE EXPRESSION

Different durations of mammary epithelium stimulation with LTA have shown variable mRNA concentrations of immune mediators such as IL-1β, IL-8, TNF-α, CXCL6, and β-defensin. A rapid increase was observed within 2–4 h of stimulation, followed by a return to baseline levels after 8–16 h, and negligible concentrations by 24 h (Strandberg et al., 2005). Acute mastitis, where exposure time to *S. aureus* is relatively short, is characterised by higher concentrations of IL-12 and IFN-γ for significantly longer time periods, reflecting its clinical nature (Bannerman et al., 2004). A dose dependent effect was demonstrated for LPS, wherein milk genes were supressed at its lower doses and immune genes were expressed at higher doses, owing to changes in methylation patterns at the corresponding promoter regions (Chen et al., 2019). Examples of the lactation-related genes reported were *ACACA*, *ACSS2*, and *S6K1* (Chen et al., 2019).

LEUCOCYTES AND DNA METHYLATION

Macrophages, neutrophils, and lymphocytes provide the next layer of protection after pathogen entry through physical barriers. They harbour PRRs on their surface and activate both innate and acquired immune responses (Sordillo et al., 1997; Soltys & Quinn, 1999). Healthy mammary tissue mainly contains macrophages, and after recognition of pathogens, these macrophages release proinflammatory cytokines leading to recruitment of neutrophils (Alhussien et al., 2021; Rainard, 2003). Neutrophils are important for defence in acute as well as chronic mastitis (Rainard & Riollet, 2003), as they act as the first line of defence and eliminate pathogens by phagocytosis (Paape et al., 2003). Level of neutrophils in the blood of cows in the puerperal period is highly heritable and is associated with susceptibility to clinical mastitis (Burvenich et al., 2000; König & May, 2019). Therefore, a decrease in neutrophil count after parturition is associated with higher incidences of mastitis (Yamaguchi et al., 1999). Also, a major constituent of neutrophil secondary granules, lactoferrin (an iron binding protein), exhibits bacteriostatic activity against *S. aureus* (Bennett & Kokocinski, 1978; Diarra et al., 2002). DNA methylation has been associated with lactoferrin expression (Grant et al., 1999), which further opens the perspectives of studying the possible immune context vis-à-vis mastitis.

Unlike neutrophils, the lymphocytes are recruited later in infection, and their interaction with *S. aureus* becomes crucial in determining the severity and outcome of the disease (Rainard et al., 2022; Ziegler et al., 2011). These include T cells, B cells, and natural killer cells. Differential DNA methylation has been observed in peripheral blood lymphocytes for associated genes involved in inflammation, namely *NRG1* (hypomethylated), *MST1* (hypomethylated), and *NAT9* (hypermethylated) during *S. aureus* subclinical mastitis, making them suitable epigenetic candidate genes (Song et al., 2016). *NRG1* (*neuregulin 1*), an epidermal growth factor, interacts with ErbB receptors in mammary gland and promotes anti-inflammatory macrophage (M2) responses (Zhao et al., 2011). In phagocytes, the *MST1* (*mammalian sterile 20-like kinase 1*) enhances the production of bactericidal reactive oxygen species (Geng et al., 2015), whereas *NAT9* (*N-acetyltransferase 9*) is known to exhibit antiviral effects.

COMPLEMENT AND DNA METHYLATION

The complement system is a part of humoral innate immune system, which includes a group of proteins that destroy the membranes of invading microorganisms and also enhance phagocytosis through opsonisation (Janeway Jr. et al., 2001). *S. aureus* is susceptible to the opsonising activity of complement proteins, but resistant to their lytic action (Laarman et al., 2011; Rooijakkers et al., 2007). This resistance is achieved by the interaction of bacterial ClfA with host complement regulator factor 1, resulting in cleavage of C3b (Hair et al., 2010). C3b is a critical component in the complement cascade. In absence of a functional C3b, formation of membrane attack complex is therefore inhibited, leading to suppressed lytic action of the complement. This mechanism of immune evasion by *S. aureus* was reported coinciding with promoter hypermethylation in the *C3* gene of Indian buffalo, revealing contributions from epigenetic regulation (Nayan et al., 2022).

IMMUNE EVASION AND EPIGENETIC MODIFICATIONS BY *S. AUREUS*

S. aureus employs multiple mechanisms to evade immune responses in host: (1) secretion of toxins and immune

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

modulators that impair host white blood cells; (2) production of protein A, which interferes with opsonisation and phagocytosis; and (3) biofilm formation, which protects the bacteria from phagocytosis and antibiotics.

Toxins and immune modulators: The organism secretes toxins and immune modulators that impair host white blood cells. Exotoxins such as TSST-1, staphylococcal enterotoxins, and staphylococcal enterotoxin-like proteins play an important role in pathogenesis within the mammary gland by acting as superantigens leading to a massive cytokine release (Fitzgerald et al., 2001; Tollersrud et al., 2006). Enterotoxins M and H cause inflammation, necrosis, and apoptosis of mammary epithelium (Liu et al., 2014; Zhao et al., 2020). Leucocidins help *S. aureus* escape the immune system by disrupting white blood cell function (Alonzo III & Torres, 2014). Tissue necrosis is caused by the α and β haemolysins (Cifrian et al., 1996; von Hoven et al., 2016). All these chemicals also increase cell permeability, exert lymphotoxic effects and lyse red blood cells for iron acquisition (Huseby et al., 2007). Research on *S. aureus* induced nasal polyps has shown enterotoxin B to influence DNA methylation patterns in the nasal tissue (Pérez-Novo et al., 2013), and its impact on methylation status in udder tissue warrants further investigation. The *TERT* (*telomerase reverse transcriptase*) gene, despite remaining hypermethylated during both coagulase-negative and coagulase-positive staphylococcal infections, showed downregulation in the latter case only (Ząbek et al., 2020). This suggests a direct role of bacterial exotoxins on the host signalling pathways, and that these bacterial factors can override methylation-based repression under certain conditions.

Protein A: Staphylococcal protein A is a surface protein found in the bacterial cell wall, and exhibits anti-opsonic properties by preventing FcR-mediated phagocytosis (Becker et al., 2014; Forsgren & Sjöquist, 1966).

Biofilm formation: In a biofilm, many layers of bacterial cells are embedded in the extracellular matrix, which adheres to biological surfaces and is made up of polysaccharide intracellular adhesions mainly (Arciola et al., 2015; Formosa-Dague et al., 2016). Attachment, microcolony formation, maturation, and detachment are the stages involved in development of biofilm (Pereyra et al., 2016). Extracellular matrix of this stratified structure acts as a barrier, providing resistance to antibiotics, and also helps in evasion of immune response, making it significant for pathogenicity of *S. aureus*. Presence of biofilm has been described in udder of cows suffering from intramammary infection (Schönborn & Krömker, 2016).

The microbe also uses several other immune evasion mechanisms in addition to the abovementioned. For example, hypomethylation of the anti-inflammatory cytokine *IL10* contributes to immune suppression, leading to chronic and subclinical mastitis (Wang et al., 2020). Although direct evidence linking various bacterial factors to DNA methylation changes in host is limited, studies on other bacterial components (Qin et al., 2021) also point to the possibility of elucidating specific effects of these elements.

INVOLVEMENT OF NON-GENIC REGIONS

Interestingly, in a study, DNA hypermethylation was observed in repeat elements LINE-1 and tRNA-derived SINE, which correlated with positive gene expression changes in immune-related genes *PADI4* and *CCAR1* respectively (Wang et al., 2024). Transposon *MTD*, member of the long terminal repeat retrotransposon family showed hypomethylation in *S. aureus* subclinical mastitis cases in the same study (Wang et al., 2024). These observations suggest that certain non-genic elements can also influence gene transcription in unexpected ways, possibly by acting as alternative regulatory elements, and can be explored in future studies. Important genes reported with DNA methylation changes in *S. aureus* -infected animals have been listed in Table 1.

GAPS IN RESEARCH

While most of the current research has focused on bulk tissue analysis, cell-specific studies on macrophages, neutrophils, etc. in addition to mammary epithelial cells, can provide more precise insights into immune regulation through DNA methylation alterations. Since the variable nature of epigenetics through time is well known, longitudinal studies can be designed to assess the long-term effects. Differences in DNA methylation responses among dairy breeds remain unexplored so far, and an understanding of these differences can be crucial for selective breeding strategies in future. A multitude of staphylococcal metabolites interact with the host's immune system to influence the outcome of infection, and more of these can be inspected in the epigenetic aspect. A comprehensive understanding of immune regulation can be developed through large-scale epigenome-wide association studies that link DNA methylation profiles with mastitis susceptibility in dairy animals.

APPLICATIONS AND POTENTIAL FUTURE RESEARCH AREAS

Based on findings from different studies, important DNA methylation markers can be selected and used as epigenetic biomarkers for early detection of *S. aureus* mastitis. Information on methylation patterns in immune-related genomic regions can assist in selection of animals with natural resistance to *S. aureus* infections. Targeted epigenetic immunotherapy to alter the immune system epigenome has been used successfully for diseases such as cancer in humans, and it can be applied for mastitis

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Genes with differential expression associated with DNA methylation changes in *Staphylococcus aureus* mastitis.

Gene	Function	DNA methylation	Gene expression	References
ACTN1	Controls the organisation of the actin cytoskeleton	Hypermethylation	Downregulated	Wang et al. (2024)
BOLA-DOB	Antigen presentation and heat resistance	Hypomethylation	Upregulated	Wang et al. (2024)
BTK	B-cell development and maturation	Hypomethylation	Upregulated	Wang et al. (2020)
C3[a]	Complement component	Hypermethylation	Downregulated	Nayan et al. (2022)
CD4	Co-receptor for T-cell receptor during antigen recognition	Hypermethylation	Downregulated	Wang et al. (2024)
CLDN3	Forms tight junctions between cells	Hypermethylation	Downregulated	Wang et al. (2024)
CLDN4	Increases the complexity of tight junctions	Hypermethylation	Downregulated	Wang et al. (2024)
CPAMD8[a]	Protease inhibitor	Hypermethylation	Downregulated	Nayan et al. (2022)
CSF2RB[a]	Common receptor β chain of cytokines and chemokines such as GMCSF, IL3, IL5	Hypermethylation	Downregulated	Nayan et al. (2022)
CSN1S1	Encodes αS1-casein milk protein	Hypermethylation	Downregulated	Wang et al. (2024)
CXADR	Cell adhesion, viral reception, and signalling	Hypermethylation	Downregulated	Wang et al. (2024)
CXCL1	Neutrophil chemotaxis, angiogenesis	Hypomethylation	Upregulated	Wu et al. (2020)
CXCL17	Angiogenesis, neutrophil traffiking, antimicrobial activity	Hypermethylation	Downregulated	Wang et al. (2024)
CXCL6	Neutrophil chemotaxis, wound healing, tissue repair	Hypomethylation	Upregulated	Wu et al. (2020)
CXCR1	Induces inflammation	Hypermethylation	Downregulated	Wang et al. (2020, 2024)
EGR1	Regulates cell proliferation and apoptosis	Hypermethylation	Downregulated	Wang et al. (2024)
FGF1	Wound healing and cell proliferation	Hypermethylation	Downregulated	Wang et al. (2024)
FGF2	Tissue repair, wound healing	Hypermethylation	Downregulated	Wang et al. (2024)
IL10	Anti-inflammatory cytokine	Hypomethylation	Upregulated	Wang et al. (2020)
IL17	Neutrophil mobilisation and activation	Hypomethylation	Upregulated	Wang et al. (2020)
IL1R2	IL1 receptor, anti-inflammatory effect	Hypermethylation	Downregulated	Wang et al. (2020)
IL-1β	Inflammation, fever	Hypomethylation	Upregulated	Wu et al. (2020) and Wang et al. (2024)
IL2RA	Regulates T-cell function	Hypomethylation	Upregulated	Wang et al. (2024)
IL-6	Inflammation, haematopoiesis	Hypomethylation	Upregulated	Wu et al. (2020)
IL6R	IL6 receptor. Induces proinflammatory response	Hypermethylation	Downregulated	Zhang et al. (2018) and Wang et al. (2024)
IL-8	Neutrophil chemotaxis, angiogenesis	Hypomethylation	Upregulated	Wu et al. (2020)
IRF2	Development and maturation of natural killer cells	Hypermethylation	Downregulated	Wang et al. (2024)
KLF4	Regulates cell proliferation, differentiation, and apoptosis	Hypermethylation	Downregulated	Wang et al. (2024)
LINE-1[b]	Retrotransposon	Hypermethylation	PADI4 upregulated	Wang et al. (2024)

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 (Continued)

Gene	Function	DNA methylation	Gene expression	References
MEF2A	T-cell activation, muscle development, cardiovascular function, and neuronal activity	Hypomethylation	Upregulated	Wang et al. (2024)
MST1	Anti-inflammatory molecule (reduces IL1β, IL6, TNFα)	Hypomethylation	Upregulated	Song et al. (2016)
NAT9	N-acetyltransferase-antiviral effect	Hypermethylation	Downregulated	Song et al. (2016)
NCF1	Promotes neutrophil activity and phagocytosis	Hypomethylation	Upregulated	Wang et al. (2024)
Nckap5[c]	Immune cell apoptosis	Hypomethylation	Upregulated	Wang et al. (2019, 2024)
NFKBIA	Negative regulator of inflammation	Hypermethylation	Downregulated	Wang et al. (2020)
NRG1	Anti-inflammatory molecule	Hypermethylation	Downregulated	Song et al. (2016)
PZP[a]	Protease inhibition; Th-1 immune response inhibition	Hypermethylation	Downregulated	Nayan et al. (2022)
RAC2	Regulates secretion, phagocytosis, and cell polarisation	Hypomethylation	Upregulated	Wang et al. (2024)
SLC9A3R1	Sodium/hydrogen exchanger regulatory cofactor (NHE-RF1)	Hypermethylation	Downregulated	Wang et al. (2024)
SP4	Enables transcription factor activity	Hypermethylation	Downregulated	Wang et al. (2024)
SPRY2	Regulates cell proliferation, differentiation, and survival	Hypermethylation	Downregulated	Wang et al. (2024)
SYK	Encodes tyrosine kinase-B cell development, cellular adhesion, innate immune recognition	Hypomethylation	Upregulated	Wang et al. (2024)
TERT	Maintains telomerase, promotes cell division	Hypermethylation	Upregulated/downregulated	Ząbek et al. (2020)
TGFB2	Regulates cell growth, differentiation, role in wound healing	Hypermethylation	Downregulated	Wang et al. (2024)
TNF-a	Pro-inflammatory cytokine	Hypermethylation	Downregulated	Wang et al. (2020)
TNFSF8	Activation and differentiation of T and B cells	Hypomethylation	Upregulated	Wang et al. (2020)
Transposon MTD[b,c]	Acts as long terminal repeat-retrotransposons	Hypomethylation	Upregulated	Wang et al. (2019)
tRNA-derived SINE[c]	Non-coding	Hypermethylation	CCAR1 upregulated	Wang et al. (2024)

[a]Study on water buffalo.
[b]Study in mouse model.
[c]Non-genic regions.

management on similar lines. Site-specific epigenetic modifications in DNA methylation patterns can be carried out using systems such as CRISPR to enhance mastitis resistance.

CONCLUSION

Over the years, the dairy industry has continued to face persistent challenge to production in the form of mastitis, accounting for substantial financial losses, and compromised animal welfare, product quality and quantity, along with public health concerns. *S. aureus*-induced mastitis is particularly a menace for the farmer due to concerns of immune evasion, AMR, and persistent infections. Traditional control strategies of improving management and genetic selection have been met with limited success due to the aforementioned factors as well as the low heritability of mastitis susceptibility trait. Therefore, supplementing these with information from epigenetic markers such as DNA methylation, becomes important in disease control. *S. aureus* modulates host immune responses and functioning through DNA methylation of many immune genes complementing its virulence factors, leading to persistence of the infection. Hypomethylation of promoter regions in immunosuppressive genes such as *IL10*, leading to their upregulation and, in contrast, hypermethylation in

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

proinflammatory *CXR1*, *TNF-α*, *IL6R*, *C3*, etc. causing their downregulation explains the lowering of immune responses caused during *S. aureus* mastitis. Hence, these genes can potentially act as epigenetic biomarkers for the disease. These biomarkers can enable early disease detection as well as selection of mastitis resilient animals. Application of epigenome-wide association studies, and use of epigenetic immunotherapy and targeted interventions through CRISPR have prospective of developing novel mastitis control strategies. Integrating epigenetic insights in genomic selection programmes can assure enhancement of the resilience of dairy animals against *S. aureus* mastitis in the future.

ACKNOWLEDGEMENTS
The authors sincerely acknowledge the Director and Joint Director (Research), ICAR-Indian Veterinary Research Institute (IVRI), Izatnagar for providing their invaluable support and necessary infrastructural facilities for this review.

CONFLICT OF INTEREST STATEMENT
The authors declare that there are no conflicts of interest associated with this study.

DATA AVAILABILITY STATEMENT
No new data were generated or analysed as part of this review.

ORCID
Sheikh Firdous Ahmad https://orcid.org/0000-0002-7114-0882
Ashwni Kumar Pandey https://orcid.org/0009-0000-1390-9983

REFERENCES
Aghamohammadi, M., Haine, D., Kelton, D.F., Barkema, H.W., Hogeveen, H., Keefe, G.P. et al. (2018) Herd-level mastitis-associated costs on Canadian dairy farms. *Frontiers in Veterinary Science*, 5, 355975.

Aitken, S.L., Corl, C.M. & Sordillo, L.M. (2011) Immunopathology of mastitis: insights into disease recognition and resolution. *Journal of Mammary Gland Biology and Neoplasia*, 16, 291–304.

Alhussien, M.N., Panda, B.S. & Dang, A.K. (2021) A comparative study on changes in total and differential milk cell counts, activity, and expression of milk phagocytes of healthy and mastitic indigenous sahiwal cows. *Frontiers in Veterinary Science*, 8, 670811.

Almeida, R.A., Matthews, K.R., Cifrian, E., Guidry, A.J. & Oliver, S.P. (1996) *Staphylococcus aureus* invasion of bovine mammary epithelial cells. *Journal of Dairy Science*, 79(6), 1021–1026.

Alonzo, F., III & Torres, V.J. (2014) The bicomponent pore-forming leucocidins of *Staphylococcus aureus*. *Microbiology and Molecular Biology Reviews*, 78(2), 199–230.

Arciola, C.R., Campoccia, D., Ravaioli, S. & Montanaro, L. (2015) Polysaccharide intercellular adhesin in biofilm: structural and regulatory aspects. *Frontiers in Cellular and Infection Microbiology*, 5, 7.

Asfaw, M. & Negash, A. (2017) Review on impact of bovine mastitis in dairy production. *Advances in Biology Research*, 11(3), 126–131.

Ashraf, A. & Imran, M. (2020) Causes, types, etiological agents, prevalence, diagnosis, treatment, prevention, effects on human health and future aspects of bovine mastitis. *Animal Health Research Reviews*, 21(1), 36–49.

Azooz, M.F., El-Wakeel, S.A. & Yousef, H.M. (2020) Financial and economic analyses of the impact of cattle mastitis on the profitability of Egyptian dairy farms. *Veterinary World*, 13(9), 1750–1759.

Bannerman, D.D., Paape, M.J., Lee, J.W., Zhao, X., Hope, J.C. & Rainard, P. (2004) *Escherichia coli* and *Staphylococcus aureus* elicit differential innate immune responses following intramammary infection. *Clinical and Vaccine Immunology*, 11(3), 463–472.

Barkema, H.W., Schukken, Y.H. & Zadoks, R.N. (2006) Invited review: the role of cow, pathogen, and treatment regimen in the therapeutic success of bovine *Staphylococcus aureus* mastitis. *Journal of Dairy Science*, 89(6), 1877–1895.

Barkema, H.W., Schukken, Y.H., Lam, T.J.G.M., Beiboer, M.L., Wilmink, H., Benedictus, G. et al. (1998) Incidence of clinical mastitis in dairy herds grouped in three categories by bulk milk somatic cell counts. *Journal of Dairy Science*, 81(2), 411–419.

Bathla, S., Sindhu, A., Kumar, S., Dubey, S.K., Pattnaik, S., Rawat, P. et al. (2020) Tandem Mass Tag (TMT)-based quantitative proteomics reveals potential targets associated with onset of subclinical mastitis in cows. *Scientific Reports*, 10(1), 9321.

Becker, S., Frankel, M.B., Schneewind, O. & Missiakas, D. (2014) Release of protein a from the cell wall of *Staphylococcus aureus*. *Proceedings of the National Academy of Sciences of the United States of America*, 111(4), 1574–1579.

Benić, M., Maćešić, N., Cvetnić, L., Habrun, B., Cvetnić, Ž., Turk, R. et al. (2018) Bovine mastitis: a persistent and evolving problem requiring novel approaches for its control-a review. *Veterinarski Arhiv*, 88(4), 535–557.

Bennett, R.M. & Kokocinski, T. (1978) Lactoferrin content of peripheral blood cells. *British Journal of Haematology*, 39(4), 509–521.

Bernstein, B.E., Meissner, A. & Lander, E.S. (2007) The mammalian epigenome. *Cell*, 128(4), 669–681.

Bobbo, T., Penasa, M. & Cassandro, M. (2019) Genetic aspects of milk differential somatic cell count in Holstein cows: a preliminary analysis. *Journal of Dairy Science*, 102(5), 4275–4279.

Bogni, C., Odierno, L., Raspanti, C., Giraudo, J., Larriestra, A., Reinoso, E. et al. (2011) War against mastitis: current concepts on controlling bovine mastitis pathogens. In: Méndez-Vilas, A. (Ed.) *Science against microbial pathogens: communicating current research and technological advances*. Singapore: World Scientific, pp. 483–494.

Bouquet, A. & Juga, J. (2013) Integrating genomic selection into dairy cattle breeding programmes: a review. *Animal*, 7(5), 705–713.

Bouzeraa, L., Martin, H., Plessis, C., Dufour, P., Marques, J.C., Moore, S. et al. (2024) Decoding epigenetic markers: implications of traits and genes through DNA methylation in resilience and susceptibility to mastitis in dairy cows. *Epigenetics*, 19(1), 2391602.

Burvenich, C., Detilleux, J., Paape, M.J. & Massart-Leën, A.M. (2000) Physiological and genetic factors that influence the cow resistance to mastitis, especially during early lactation. *Flemish Veterinary Journal*, 9–20.

Campos, B., Pickering, A.C., Rocha, L.S., Aguilar, A.P., Fabres-Klein, M.H., de Oliveira Mendes, T.A. et al. (2022) Diversity and pathogenesis of *Staphylococcus aureus* from bovine mastitis: current understanding and future perspectives. *BMC Veterinary Research*, 18(1), 115.

Chandrasekaran, D., Venkatesan, P., Tirumurugaan, K.G., Nambi, A.P., Thirunavukkarasu, P.S., Kumanan, K. et al. (2014) Pattern of antibiotic resistant mastitis in dairy cows. *Veterinary World*, 7(6), 389–394.

Chen, J., Wu, Y., Sun, Y., Dong, X., Wang, Z., Zhang, Z. et al. (2019) Bacterial lipopolysaccharide induced alterations of genome-wide DNA methylation and promoter methylation of lactation-related genes in bovine mammary epithelial cells. *Toxins*, 11(5), 298.

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Cheng, J., Zhang, J., Han, B., Barkema, H.W., Cobo, E.R., Kastelic, J.P. et al. (2020) *Klebsiella pneumoniae* isolated from bovine mastitis is cytopathogenic for bovine mammary epithelial cells. *Journal of Dairy Science*, 103(4), 3493–3504.

Cheng, R., Wang, Q. & Wei, L. (2022) Income growth, employment structure transition and the rise of modern markets: the impact of urbanization on residents' consumption of dairy products in China. *PLoS One*, 17(4), e0267006.

Cheng, W.N. & Han, S.G. (2020) Bovine mastitis: risk factors, therapeutic strategies, and alternative treatments—a review. *Asian-Australasian Journal of Animal Sciences*, 33(11), 1699–1713.

Cifrian, E., Guidry, A.J., Bramley, A.J., Norcross, N.L., Bastida-Corcuera, F.D. & Marquardt, W.W. (1996) Effect of staphylococcal β toxin on the cytotoxicity, proliferation and adherence of *Staphylococcus aureus* to bovine mammary epithelial cells. *Veterinary Microbiology*, 48(3–4), 187–198.

Cobirka, M., Tancin, V. & Slama, P. (2020) Epidemiology and classification of mastitis. *Animals*, 10(12), 2212.

Dalanezi, F.M., Joaquim, S.F., Guimarães, F.F., Guerra, S.T., Lopes, B.C., Schmidt, E.M.S. et al. (2020) Influence of pathogens causing clinical mastitis on reproductive variables of dairy cows. *Journal of Dairy Science*, 103(4), 3648–3655.

de Haas, Y., Barkema, H.W. & Veerkamp, R.F. (2002) Genetic parameters of pathogen-specific incidence of clinical mastitis in dairy cows. *Animal Science*, 74(2), 233–242.

De Jong, N.W., Van Kessel, K.P. & Van Strijp, J.A. (2019) Immune evasion by *Staphylococcus aureus*. *Microbiology Spectrum*, 7(2), 10–1128.

De Vliegher, S., Fox, L.K., Piepers, S., McDougall, S. & Barkema, H.W. (2012) Invited review: mastitis in dairy heifers: nature of the disease, potential impact, prevention, and control. *Journal of Dairy Science*, 95(3), 1025–1040.

Dego, O.K. (2020) Bovine mastitis: part I. In: *Animal reproduction in veterinary medicine*. IntechOpen, p. 149.

Detilleux, J., Kastelic, J.P. & Barkema, H.W. (2015) Mediation analysis to estimate direct and indirect milk losses due to clinical mastitis in dairy cattle. *Preventive Veterinary Medicine*, 118(4), 449–456.

Diarra, M.S., Petitclerc, D. & Lacasse, P. (2002) Effect of lactoferrin in combination with penicillin on the morphology and the physiology of *Staphylococcus aureus* isolated from bovine mastitis. *Journal of Dairy Science*, 85(5), 1141–1149.

Dinarello, C.A. (2000) Proinflammatory cytokines. *Chest*, 118(2), 503–508.

Dong, Y., An, D., Wang, J., Liu, H., Zhang, Q., Zhao, J. et al. (2021) Effect of DNA methylation on LPS-induced expression of tumour necrosis factor alpha (TNF-α) in bovine mammary epithelial cells. *Indian Journal of Animal Research*, 55(9), 1079–1084.

Dufour, S., Dohoo, I.R., Barkema, H.W., Descôteaux, L., Devries, T.J., Reyher, K.K. et al. (2012) Epidemiology of coagulase-negative staphylococci intramammary infection in dairy cattle and the effect of bacteriological culture misclassification. *Journal of Dairy Science*, 95(6), 3110–3124.

Eckel, E.F. & Ametaj, B.N. (2016) Invited review: role of bacterial endotoxins in the etiopathogenesis of periparturient diseases of transition dairy cows. *Journal of Dairy Science*, 99(8), 5967–5990.

Engler, C., Renna, M.S., Beccaria, C., Silvestrini, P., Pirola, S.I., Pereyra, E.A. et al. (2022) Differential immune response to two *Staphylococcus aureus* strains with distinct adaptation genotypes after experimental intramammary infection of dairy cows. *Microbial Pathogenesis*, 172, 105789.

Erskine, R.J., Walker, R.D., Bolin, C.A., Bartlett, P.C. & White, D.G. (2002) Trends in antibacterial susceptibility of mastitis pathogens during a seven-year period. *Journal of Dairy Science*, 85(5), 1111–1118.

Fitzgerald, J.R., Monday, S.R., Foster, T.J., Bohach, G.A., Hartigan, P.J., Meaney, W.J. et al. (2001) Characterization of a putative pathogenicity island from bovine *Staphylococcus aureus* encoding multiple superantigens. *Journal of Bacteriology*, 183(1), 63–70.

Formosa-Dague, C., Feuillie, C., Beaussart, A., Derclaye, S., Kucharikova, S., Lasa, I. et al. (2016) Sticky matrix: adhesion mechanism of the staphylococcal polysaccharide intercellular adhesin. *ACS Nano*, 10(3), 3443–3452.

Forsgren, A. & Sjöquist, J. (1966) "Protein a" from *S. aureus*: I. Pseudo-immune reaction with human γ-globulin. *The Journal of Immunology*, 97(6), 822–827.

García-Álvarez, L., Holden, M.T., Lindsay, H., Webb, C.R., Brown, D.F., Curran, M.D. et al. (2011) Meticillin-resistant *Staphylococcus aureus* with a novel mecA homologue in human and bovine populations in the UK and Denmark: a descriptive study. *The Lancet Infectious Diseases*, 11(8), 595–603.

Geng, J., Sun, X., Wang, P., Zhang, S., Wang, X., Wu, H. et al. (2015) Kinases Mst1 and Mst2 positively regulate phagocytic induction of reactive oxygen species and bactericidal activity. *Nature Immunology*, 16(11), 1142–1152.

Gogoi-Tiwari, J., Williams, V., Waryah, C.B., Costantino, P., Al-Salami, H., Mathavan, S. et al. (2017) Mammary gland pathology subsequent to acute infection with strong versus weak biofilm forming *Staphylococcus aureus* bovine mastitis isolates: a pilot study using non-invasive mouse mastitis model. *PLoS One*, 12(1), e0170668.

Grant, D.J., Shi, H. & Teng, C.T. (1999) Tissue and site-specific methylation correlates with expression of the mouse lactoferrin gene. *Journal of Molecular Endocrinology*, 23(1), 45–55.

Günther, J. & Seyfert, H.M. (2018) The first line of defence: insights into mechanisms and relevance of phagocytosis in epithelial cells. In: *Seminars in immunopathology*, Vol. 40. Berlin/Heidelberg: Springer Berlin Heidelberg, pp. 555–565.

Hair, P.S., Echague, C.G., Sholl, A.M., Watkins, J.A., Geoghegan, J.A., Foster, T.J. et al. (2010) Clumping factor a interaction with complement factor I increases C3b cleavage on the bacterial surface of *Staphylococcus aureus* and decreases complement-mediated phagocytosis. *Infection and Immunity*, 78(4), 1717–1727.

Halasa, T., Huijps, K., Østerås, O. & Hogeveen, H. (2007) Economic effects of bovine mastitis and mastitis management: a review. *Veterinary Quarterly*, 29(1), 18–31.

Heikkilä, A.M., Liski, E., Pyörälä, S. & Taponen, S. (2018) Pathogen-specific production losses in bovine mastitis. *Journal of Dairy Science*, 101(10), 9493–9504.

Hernandez, A., Quinton, M., Miglior, F. & Mallard, B.A. (2006) *Genetic parameters of dairy cattle immune response traits*.

Ho, D.H. & Burggren, W.W. (2010) Epigenetics and transgenerational transfer: a physiological perspective. *Journal of Experimental Biology*, 213(1), 3–16.

Holmberg, M., Fikse, W.F., Andersson-Eklund, L., Artursson, K. & Lundén, A. (2012) Genetic analyses of pathogen-specific mastitis. *Journal of Animal Breeding and Genetics*, 129(2), 129–137.

Holmes, M.A. & Zadoks, R.N. (2011) Methicillin resistant *S. aureus* in human and bovine mastitis. *Journal of Mammary Gland Biology and Neoplasia*, 16, 373–382.

Huseby, M., Shi, K., Brown, C.K., Digre, J., Mengistu, F., Seo, K.S. et al. (2007) Structure and biological activities of beta toxin from *Staphylococcus aureus*. *Journal of Bacteriology*, 189(23), 8719–8726.

Ibeagha-Awemu, E.M. & Khatib, H. (2017) Epigenetics of livestock breeding. In: *Handbook of epigenetics*. Academic Press, pp. 441–463.

Ibeagha-Awemu, E.M. & Zhao, X. (2015) Epigenetic marks: regulators of livestock phenotypes and conceivable sources of missing variation in livestock improvement programs. *Frontiers in genetics*, 6, 302.

Ibeagha-Awemu, E.M. & Yu, Y. (2021) Consequence of epigenetic processes on animal health and productivity: is additional level of regulation of relevance? *Animal Frontiers*, 11(6), 7–18.

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Illingworth, R.S. & Bird, A.P. (2009) CpG islands – 'a rough guide'. *FEBS Letters*, 583(11), 1713–1720.

Ivanova, E., Le Guillou, S., Hue-Beauvais, C. & Le Provost, F. (2021) Epigenetics: new insights into mammary gland biology. *Genes*, 12(2), 231.

Jamali, H., Barkema, H.W., Jacques, M., Lavallée-Bourget, E.M., Malouin, F., Saini, V. et al. (2018) Invited review: incidence, risk factors, and effects of clinical mastitis recurrence in dairy cows. *Journal of Dairy Science*, 101(6), 4729–4746.

James, S.J., Cutler, P., Melnyk, S., Jernigan, S., Janak, L., Gaylor, D.W. et al. (2004) Metabolic biomarkers of increased oxidative stress and impaired methylation capacity in children with autism. *The American Journal of Clinical Nutrition*, 80(6), 1611–1617.

Janeway, C.A., Jr., Travers, P., Walport, M. & Shlomchik, M.J. (2001) The complement system and innate immunity. In: *Immunobiology: the immune system in health and disease*, 5th edition. Garland Science.

Jones, G.M. & Bailey, T.L. (2009) *Understanding the basics of mastitis*.

Ju, Z., Jiang, Q., Wang, J., Wang, X., Yang, C., Sun, Y. et al. (2020) Genome-wide methylation and transcriptome of blood neutrophils reveal the roles of DNA methylation in affecting transcription of protein-coding genes and miRNAs in *E. coli*-infected mastitis cows. *BMC Genomics*, 21, 1–14.

Kibebew, K. (2017) Bovine mastitis: a review of causes and epidemiological point of view. *Journal of Biology, Agriculture and Healthcare*, 7(2), 1–14.

König, S. & May, K. (2019) Invited review: phenotyping strategies and quantitative-genetic background of resistance, tolerance and resilience associated traits in dairy cattle. *Animal*, 13(5), 897–908.

Krishnamoorthy, P., Suresh, K.P., Jayamma, K.S., Shome, B.R., Patil, S.S. & Amachawadi, R.G. (2021) An understanding of the global status of major bacterial pathogens of milk concerning bovine mastitis: a systematic review and meta-analysis (Scientometrics). *Pathogens*, 10(5), 545.

Kumar, N., Manimaran, A., Kumaresan, A., Jeyakumar, S., Sreela, L., Mooventhan, P. et al. (2017) Mastitis effects on reproductive performance in dairy cattle: a review. *Tropical Animal Health and Production*, 49, 663–673.

Laarman, A.J., Ruyken, M., Malone, C.L., van Strijp, J.A., Horswill, A.R. & Rooijakkers, S.H. (2011) *Staphylococcus aureus* metalloprotease aureolysin cleaves complement C3 to mediate immune evasion. *The Journal of Immunology*, 186(11), 6445–6453.

Lahouassa, H., Moussay, E., Rainard, P. & Riollet, C. (2007) Differential cytokine and chemokine responses of bovine mammary epithelial cells to *Staphylococcus aureus* and *Escherichia coli*. *Cytokine*, 38(1), 12–21.

Langevin, S.M. & Kelsey, K.T. (2013) The fate is not always written in the genes: epigenomics in epidemiologic studies. *Environmental and Molecular Mutagenesis*, 54(7), 533–541.

Lerminiaux, N.A. & Cameron, A.D. (2019) Horizontal transfer of antibiotic resistance genes in clinical environments. *Canadian Journal of Microbiology*, 65(1), 34–44.

Levison, L.J., Miller-Cushon, E.K., Tucker, A.L., Bergeron, R., Leslie, K.E., Barkema, H.W. et al. (2016) Incidence rate of pathogen-specific clinical mastitis on conventional and organic Canadian dairy farms. *Journal of Dairy Science*, 99(2), 1341–1350.

Liu, Y., Chen, W., Ali, T., Alkasir, R., Yin, J., Liu, G. et al. (2014) Staphylococcal enterotoxin H induced apoptosis of bovine mammary epithelial cells in vitro. *Toxins*, 6(12), 3552–3567.

Logue, C.J.K. & Nolan, A. (1998) *Recognising and controlling pain and inflammation in mastitis*.

Lush, J.L. (1950) Inheritance of susceptibility to mastitis. *Journal of Dairy Science*, 33(2), 121–125.

Maddela, P., Makwana, P.M., Parmar, S.M., Kalyani, I.H., Patel, D.R. & Parasana, D.K. (2024) Cultural and biochemical characterization of *Staphylococcus aureus* isolates from bovine clinical and subclinical mastitis. *Indian Journal of Veterinary Sciences & Biotechnology*, 20(3).

Mallard, B.A., Emam, M., Paibomesai, M., Thompson-Crispi, K. & Wagter-Lesperance, L. (2015) Genetic selection of cattle for improved immunity and health. *Japanese Journal of Veterinary Research*, 63(Supplement 1), S37–S44.

Mao, Y.J., Zhu, X.R., Li, R., Chen, D., Xin, S.Y., Zhu, Y.H. et al. (2015) Methylation analysis of CXCR1 in mammary gland tissue of cows with mastitis induced by *Staphylococcus aureus*. *Genetics and Molecular Research*, 14, 12606–12615.

Middleton, J.R. (2008) *Staphylococcus aureus* antigens and challenges in vaccine development. *Expert Review of Vaccines*, 7(6), 805–815.

Miranda, T.B. & Jones, P.A. (2007) DNA methylation: the nuts and bolts of repression. *Journal of Cellular Physiology*, 213(2), 384–390.

Morales-Ubaldo, A.L., Rivero-Perez, N., Valladares-Carranza, B., Velázquez-Ordoñez, V., Delgadillo-Ruiz, L. & Zaragoza-Bastida, A. (2023) Bovine mastitis, a worldwide impact disease: prevalence, antimicrobial resistance, and viable alternative approaches. *Veterinary and Animal Science*, 21, 100306.

Narayana, S.G., Schenkel, F., Miglior, F., Chud, T., Abdalla, E.A., Naqvi, S.A. et al. (2021) Genetic analysis of pathogen-specific intramammary infections in dairy cows. *Journal of Dairy Science*, 104(2), 1982–1992.

Nayan, V., Singh, K., Iquebal, M.A., Jaiswal, S., Bhardwaj, A., Singh, C. et al. (2022) Genome-wide DNA methylation and its effect on gene expression during subclinical mastitis in water buffalo. *Frontiers in Genetics*, 13, 828292.

Nigam, R. (2015) Incidence and pattern of antibiotic resistance of *Staphylococcus aureus* isolated from clinical and subclinical mastitis in cattle and buffaloes. *Asian Journal of Animal Sciences*, 9(3), 100–109.

Notcovich, S. (2021) *The physiology of the keratin plug formation in the teat canal of dairy cattle and its interaction with current and novel methods for prevention of intramammary infections: a thesis presented in partial fulfilment of the requirements for the degree of Doctor of Philosophy in Veterinary Science, Massey University, Turitea, Palmerston North, New Zealand* (Doctoral dissertation, Massey University).

Novick, R.P. & Geisinger, E. (2008) Quorum sensing in staphylococci. *Annual Review of Genetics*, 42(1), 541–564.

Obšteter, J., Jenko, J. & Gorjanc, G. (2021) Genomic selection for any dairy breeding program via optimized investment in phenotyping and genotyping. *Frontiers in Genetics*, 12, 637017.

Paape, M., Bannerman, D., Zhao, X. & Lee, J.W. (2003) The bovine neutrophil: structure and function in blood and milk. *Veterinary Research*, 34(5), 597–627.

Pereyra, E.A., Picech, F., Renna, M.S., Baravalle, C., Andreotti, C.S., Russi, R. et al. (2016) Detection of *Staphylococcus aureus* adhesion and biofilm-producing genes and their expression during internalization in bovine mammary epithelial cells. *Veterinary Microbiology*, 183, 69–77.

Pérez-Novo, C.A., Zhang, Y., Denil, S., Trooskens, G., De Meyer, T., Van Criekinge, W. et al. (2013) Staphylococcal enterotoxin B influences the DNA methylation pattern in nasal polyp tissue: a preliminary study. *Allergy, Asthma & Clinical Immunology*, 9, 1–5.

Petronis, A. (2010) Epigenetics as a unifying principle in the aetiology of complex traits and diseases. *Nature*, 465(7299), 721–727.

Pighetti, G.M. & Elliott, A.A. (2011) Gene polymorphisms: the keys for marker assisted selection and unraveling core regulatory pathways for mastitis resistance. *Journal of Mammary Gland Biology and Neoplasia*, 16, 421–432.

Qin, W., Scicluna, B.P. & van der Poll, T. (2021) The role of host cell DNA methylation in the immune response to bacterial infection. *Frontiers in Immunology*, 12, 696280.

Rainard, P. (2003) The complement in milk and defense of the bovine mammary gland against infections. *Veterinary Research*, 34(5), 647–670.

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Rainard, P. & Riollet, C. (2003) Mobilization of neutrophils and defense of the bovine mammary gland. *Reproduction Nutrition Development*, 43(5), 439–457.

Rainard, P. & Riollet, C. (2006) Innate immunity of the bovine mammary gland. *Veterinary Research*, 37(3), 369–400.

Rainard, P., Foucras, G. & Martins, R.P. (2022) Adaptive cell-mediated immunity in the mammary gland of dairy ruminants. *Frontiers in Veterinary Science*, 9, 854890.

Reyher, K.K., Dufour, S., Barkema, H.W., Des Côteaux, L., Devries, T.J., Dohoo, I.R. et al. (2011) The National Cohort of Dairy Farms—a data collection platform for mastitis research in Canada. *Journal of Dairy Science*, 94, 1616–1626.

Rooijakkers, S.H., Milder, F.J., Bardoel, B.W., Ruyken, M., van Strijp, J.A. & Gros, P. (2007) Staphylococcal complement inhibitor: structure and active sites. *The Journal of Immunology*, 179(5), 2989–2998.

Saidani, M., Messadi, L., Soudani, A., Daaloul-Jedidi, M., Châtre, P., Ben Chehida, F. et al. (2018) Epidemiology, antimicrobial resistance, and extended-spectrum beta-lactamase-producing Enterobacteriaceae in clinical bovine mastitis in Tunisia. *Microbial Drug Resistance*, 24(8), 1242–1248.

Schönborn, S. & Krömker, V. (2016) Detection of the biofilm component polysaccharide intercellular adhesin in *Staphylococcus aureus* infected cow udders. *Veterinary Microbiology*, 196, 126–128.

Singha, S., Koop, G., Persson, Y., Hossain, D., Scanlon, L., Derks, M. et al. (2021) Incidence, etiology, and risk factors of clinical mastitis in dairy cows under semi-tropical circumstances in Chattogram, Bangladesh. *Animals*, 11(8), 2255.

Sivakumar, R., Pranav, P.S., Annamanedi, M., Chandrapriya, S., Isloor, S., Rajendhran, J. et al. (2023) Genome sequencing and comparative genomic analysis of bovine mastitis-associated *Staphylococcus aureus* strains from India. *BMC Genomics*, 24(1), 44.

Smolenski, G.A., Cursons, R.T., Hine, B.C. & Wheeler, T.T. (2015) Keratin and S100 calcium-binding proteins are major constituents of the bovine teat canal lining. *Veterinary Research*, 46, 1–10.

Soares, B.S., Melo, D.A., Motta, C.C., Marques, V.F., Barreto, N.B., Coelho, S.M.O. et al. (2017) Characterization of virulence and antibiotic profile and agr typing of *Staphylococcus aureus* from milk of subclinical mastitis bovine in state of Rio de Janeiro. *Arquivo Brasileiro de Medicina Veterinária e Zootecnia*, 69, 843–850.

Soltys, J. & Quinn, M.T. (1999) Selective recruitment of T-cell subsets to the udder during staphylococcal and streptococcal mastitis: analysis of lymphocyte subsets and adhesion molecule expression. *Infection and Immunity*, 67(12), 6293–6302.

Song, M., He, Y., Zhou, H., Zhang, Y., Li, X. & Yu, Y. (2016) Combined analysis of DNA methylome and transcriptome reveal novel candidate genes with susceptibility to bovine *Staphylococcus aureus* subclinical mastitis. *Scientific Reports*, 6(1), 29390.

Sordillo, L.M., Shafer-Weaver, K. & DeRosa, D. (1997) Immunobiology of the mammary gland. *Journal of Dairy Science*, 80(8), 1851–1865.

Strandberg, Y., Gray, C., Vuocolo, T., Donaldson, L., Broadway, M. & Tellam, R. (2005) Lipopolysaccharide and lipoteichoic acid induce different innate immune responses in bovine mammary epithelial cells. *Cytokine*, 31(1), 72–86.

Sutra, L. & Poutrel, B. (1994) Virulence factors involved in the pathogenesis of bovine intramammary infections due to *Staphylococcus aureus*. *Journal of Medical Microbiology*, 40(2), 79–89.

Tajima, S., Suetake, I., Takeshita, K., Nakagawa, A., Kimura, H. & Song, J. (2022) Domain structure of the Dnmt1, Dnmt3a, and Dnmt3b DNA methyltransferases. In: *DNA methyltransferases-role and function*. Springer, pp. 45–68.

Tando, Y. & Matsui, Y. (2023) Inheritance of environment-induced phenotypic changes through epigenetic mechanisms. *Environmental Epigenetics*, 9(1), dvad008.

Thompson-Crispi, K.A., Sewalem, A., Miglior, F. & Mallard, B.A. (2012) Genetic parameters of adaptive immune response traits in Canadian Holsteins. *Journal of Dairy Science*, 95(1), 401–409.

Tollersrud, T., Kampen, A.H. & Kenny, K. (2006) *Staphylococcus aureus* enterotoxin D is secreted in milk and stimulates specific antibody responses in cows in the course of experimental intramammary infection. *Infection and Immunity*, 74(6), 3507–3512.

Triantaphyllopoulos, K.A., Ikonomopoulos, I. & Bannister, A.J. (2016) Epigenetics and inheritance of phenotype variation in livestock. *Epigenetics & Chromatin*, 9, 1–18.

Turner, M.D., Nedjai, B., Hurst, T. & Pennington, D.J. (2014) Cytokines and chemokines: at the crossroads of cell signalling and inflammatory disease. *Biochimica et Biophysica Acta (BBA)-Molecular Cell Research*, 1843(11), 2563–2582.

Vanselow, J., Yang, W., Herrmann, J., Zerbe, H., Schuberth, H.J., Petzl, W. et al. (2006) DNA-remethylation around a STAT5-binding enhancer in the αS1-casein promoter is associated with abrupt shutdown of αS1-casein synthesis during acute mastitis. *Journal of Molecular Endocrinology*, 37(3), 463–477.

Vitenberga-Verza, Z., Pilmane, M., Šerstņova, K., Melderis, I., Gontar, Ł., Kochański, M. et al. (2022) Identification of inflammatory and regulatory cytokines IL-1α-, IL-4-, IL-6-, IL-12-, IL-13-, IL-17A-, TNF-α-, and IFN-γ-producing cells in the milk of dairy cows with subclinical and clinical mastitis. *Pathogens*, 11(3), 372.

von Hoven, G., Rivas, A.J., Neukirch, C., Klein, S., Hamm, C., Qin, Q. et al. (2016) Dissecting the role of ADAM10 as a mediator of *Staphylococcus aureus* α-toxin action. *Biochemical Journal*, 473(13), 1929–1940.

Walawski, K. (1999) Genetic aspects of mastitis resistance in cattle. *Journal of Applied Genetics*, 40(2), 117–128.

Wang, D., Wei, Y., Shi, L., Khan, M.Z., Fan, L., Wang, Y. et al. (2019) Genome-wide DNA methylation pattern in a mouse model reveals two novel genes associated with *Staphylococcus aureus* mastitis. *Asian-Australasian Journal of Animal Sciences*, 33(2), 203.

Wang, M. & Ibeagha-Awemu, E.M. (2021) Impacts of epigenetic processes on the health and productivity of livestock. *Frontiers in Genetics*, 11, 613636.

Wang, M., Bissonnette, N., Laterrière, M., Dudemaine, P.L., Gagné, D., Roy, J.P. et al. (2024) DNA methylation haplotype block signatures responding to *Staphylococcus aureus* subclinical mastitis and association with production and health traits. *BMC Biology*, 22(1), 65.

Wang, M., Bissonnette, N., Laterrière, M., Gagné, D., Dudemaine, P.L., Roy, J.P. et al. (2023) Genome-wide DNA methylation and transcriptome integration associates DNA methylation changes with bovine subclinical mastitis caused by *Staphylococcus chromogenes*. *International Journal of Molecular Sciences*, 24(12), 10369.

Wang, M., Laterrière, M., Dudemaine, P.L., Bissonnette, N., Gagné, D. & Ibeagha-Awemu, E.M. (2022) PSVIII-B-15 potential involvement of DNA methylation of first intron in transcriptional regulation during bovine subclinical mastitis caused by *Staphylococcus aureus*. *Journal of Animal Science*, 100(Supplement_3), 310–311.

Wang, M., Liang, Y., Ibeagha-Awemu, E.M., Li, M., Zhang, H., Chen, Z. et al. (2020) Genome-wide DNA methylation analysis of mammary gland tissues from Chinese Holstein cows with *Staphylococcus aureus* induced mastitis. *Frontiers in Genetics*, 11, 550515.

Wellnitz, O. & Bruckmaier, R.M. (2012) The innate immune response of the bovine mammary gland to bacterial infection. *The Veterinary Journal*, 192(2), 148–152.

Wellnitz, O., Arnold, E.T. & Bruckmaier, R.M. (2011) Lipopolysaccharide and lipoteichoic acid induce different immune responses in the bovine mammary gland. *Journal of Dairy Science*, 94(11), 5405–5412.

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Wu, Y., Chen, J., Sun, Y., Dong, X., Wang, Z., Chen, J. et al. (2020) PGN and LTA from *Staphylococcus aureus* induced inflammation and decreased lactation through regulating DNA methylation and histone H3 acetylation in bovine mammary epithelial cells. *Toxins*, 12(4), 238.

Yamaguchi, T., Hiratsuka, M., Asai, K., Kai, K. & Kumagai, K. (1999) Differential distribution of T lymphocyte subpopulations in the bovine mammary gland during lactation. *Journal of Dairy Science*, 82(7), 1459–1464.

Yang, W., Zerbe, H., Petzl, W., Brunner, R.M., Günther, J., Draing, C. et al. (2008) Bovine TLR2 and TLR4 properly transduce signals from *Staphylococcus aureus* and *E. coli*, but *S. aureus* fails to both activate NF-κB in mammary epithelial cells and to quickly induce TNFα and interleukin-8 (CXCL8) expression in the udder. *Molecular Immunology*, 45(5), 1385–1397.

Yousefian, N., Alam, S., Ramappa, K.B., Schlecht, E. & Dittrich, C. (2024) Cows in the city: how consumer demands sustain urban dairying in the IT capital of India. *Urban Agriculture & Regional Food Systems*, 9(1), e20070.

Ząbek, T., Semik-Gurgul, E., Ropka-Molik, K., Szmatoła, T., Kawecka-Grochocka, E., Zalewska, M. et al. (2020) Locus-specific interrelations between gene expression and DNA methylation patterns in bovine mammary gland infected by coagulase-positive and coagulase-negative staphylococci. *Journal of Dairy Science*, 103(11), 10689–10695.

Zhang, Y., Wang, X., Jiang, Q., Hao, H., Ju, Z., Yang, C. et al. (2018) DNA methylation rather than single nucleotide polymorphisms regulates the production of an aberrant splice variant of IL6R in mastitic cows. *Cell Stress and Chaperones*, 23(4), 617–628.

Zhao, W., Shen, Y. & Ren, S. (2011) Endogenous expression of Neuregulin-1 (Nrg1) as a potential modulator of prolactin (PRL) secretion in GH3 cells. *Cell and Tissue Research*, 344, 313–320.

Zhao, Y., Tang, J., Yang, D., Tang, C. & Chen, J. (2020) Staphylococcal enterotoxin M induced inflammation and impairment of bovine mammary epithelial cells. *Journal of Dairy Science*, 103(9), 8350–8359.

Zi, C., Zeng, D., Ling, N., Dai, J., Xue, F., Jiang, Y. et al. (2018) An improved assay for rapid detection of viable *Staphylococcus aureus* cells by incorporating surfactant and PMA treatments in qPCR. *BMC Microbiology*, 18, 1–8.

Ziegler, C., Goldmann, O., Hobeika, E., Geffers, R., Peters, G. & Medina, E. (2011) The dynamics of T cells during persistent *Staphylococcus aureus* infection: from antigen-reactivity to in vivo anergy. *EMBO Molecular Medicine*, 3(11), 652–666.

How to cite this article: Apeksha, Todkari, A.M., Chaudhary, A., Hassan, M.M., Upreti, D., Saini, S.R. et al. (2025) Decoding DNA methylation in *Staphylococcus aureus* mastitis: Implications for immune regulation and disease resistance. *Animal Genetics*, 56, e70044. Available from: https://doi.org/10.1111/age.70044

13652052, 2025, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/age.70044 by Bangladesh Hinari NPL, Wiley Online Library on [23/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License